

Qualification Pack

Cyber Security Assistant

QP Code: NIE/SSC/Q0904

Version: 1.0

NSQF Level: 3

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NIE/SSC/Q0904: Cyber Security Assistant

Brief Job Description

The Cyber Security Assistant Qualification would provide the fundamental skillset to the enrolled trainee, pertaining to the field of Cyber Security. After course completion, the trained learner will exhibit a refined posture about the basic concepts of Cyber Security ranging from Primitive Networking, Fundamentals of Cyber Security, Operating System Introduction, Wireless, Mobile Security, OS, Web, Social engineering, and Cyber laws

Personal Attributes

A Cyber Security Assistant should possess strong attention to detail, integrity, and critical thinking skills to identify and address security threats effectively. They must be trustworthy, adaptable, and capable of handling confidential information responsibly. Strong communication, teamwork, and a commitment to continuous learning are essential in this fast-evolving field.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)
2. [NIE/ITS/N0901: Fundamentals of Network and Operating System](#)
3. [NIE/ITS/N0902: Fundamentals of Cyber Security](#)
4. [NIE/SSC/N0918: Implementation of Cyber Security Project](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	Networking and Cybersecurity
Occupation	Cyber Security
Country	India
NSQF Level	3
Credits	10
Aligned to NCO/ISCO/ISIC Code	3

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Minimum Educational Qualification & Experience	10th grade pass with NA of experience OR Grade 8 pass with 2 years of (NTC/ NAC) after 8th with NA of experience OR 8th grade pass and pursuing continuous schooling in regular school with vocational subject with NA of experience OR 8th grade pass (with experience in IT-ITeS Sector) with 2 Years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	16 Years
Last Reviewed On	NA
Next Review Date	29/03/2026
NSQC Approval Date	29/03/2023
Version	1.0
Reference code on NQR	QG-03-IT-00350-2023-V1-NIELIT
NQR Version	1

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DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

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NIE/ITS/N0901: Fundamentals of Network and Operating System

Description

Fundamentals of Network and Operating Systems encompass competencies in understanding networking fundamentals, operating system basics, network configuration and management, network services and protocols, troubleshooting, virtualization, security, documentation, and compliance within the context of network and operating system environments.

Scope

The scope covers the following :

- Networking fundamentals, configuration and management, operating system basics, virtualization, security

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Overview of OSI, TCP/IP Model and Function of layers, learn about various types of networks.
- PC2.** Knowledge about the Benefits of Networks, and components of Computer Networks and also have an insight into various topologies used in networking.
- PC3.** Explain various Transmission modes, Ethernet, and the Classification Of Transmission Media.
- PC4.** In-depth idea about various networking devices like Repeaters, Hubs, Bridges, Switches, and routers
- PC5.** Introduction to TCP/IP Protocol Model in Depth, Port Numbers, Common TCP& UDP Ports and also Some important TCP/IP protocols.
- PC6.** IP addressing, Classes, IP Addressing Scheme, Subnetting Basics alongwith How to Create Subnets, Subnet Masks, Subnetting Class C Addresses, Subnetting Class B Addresses and VLSM
- PC7.** Configure ip addresses in Windows and Linux and also have a proper introduction to RIR, registries and IANA along with attaining knowledge about TCP/IP Troubleshooting utilities, Troubleshooting IP Addressing
- PC8.** The in-depth idea about the Windows and Linux operating system and its File System
- PC9.** Working on Command Line Utilities
- PC10.** Working with Links: Working with Symbolic Links, Working with Hard Links
- PC11.** Configuring Permissions, Read, Write, and Execute:
- PC12.** Preparing the File System for ACLs
- PC13.** Configuring Administrator Tasks with sudo
- PC14.** The in-depth idea about the Linux operating system and its File System
- PC15.** Managing Users and Groups,

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** Understand the fundamentals of computer networking, including network types, devices, and data transmission concepts.
- KU2.** Gain in-depth knowledge of the OSI Reference Model and TCP/IP Protocol Suite, including their functions and inter-layer communication.
- KU3.** Learn to use key TCP/IP command-line troubleshooting utilities such as ping, ipconfig, tracert, and netstat for diagnosing network issues.
- KU4.** Acquire basic knowledge of navigating and exploring both Windows and Linux operating systems, including file systems, commands, and user interfaces.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Fundamental Networking Competency -Ability to understand and explain basic networking concepts, including the OSI and TCP/IP models.
- GS2.** Practical Troubleshooting Skills -Proficient in using command-line utilities for diagnosing and resolving TCP/IP network issues.
- GS3.** Multi-OS Literacy - Capable of navigating, exploring, and operating both Windows and Linux operating systems.
- GS4.** Analytical & Problem-Solving Abilities -Able to analyze system behavior across platforms and resolve configuration or connectivity challenges efficiently.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	100	30	-	-
PC1. Overview of OSI, TCP/IP Model and Function of layers, learn about various types of networks.	-	-	-	-
PC2. Knowledge about the Benefits of Networks, and components of Computer Networks and also have an insight into various topologies used in networking.	-	-	-	-
PC3. Explain various Transmission modes, Ethernet, and the Classification Of Transmission Media.	-	-	-	-
PC4. In-depth idea about various networking devices like Repeaters, Hubs, Bridges, Switches, and routers	-	-	-	-
PC5. Introduction to TCP/IP Protocol Model in Depth, Port Numbers, Common TCP& UDP Ports and also Some important TCP/IP protocols.	-	-	-	-
PC6. IP addressing, Classes, IP Addressing Scheme, Subnetting Basics alongwith How to Create Subnets, Subnet Masks, Subnetting Class C Addresses, Subnetting Class B Addresses and VLSM	-	-	-	-
PC7. Configure ip addresses in Windows and Linux and also have a proper introduction to RIR, registries and IANA along with attaining knowledge about TCP/IP Troubleshooting utilities, Troubleshooting IP Addressing	-	-	-	-
PC8. The in-depth idea about the Windows and Linux operating system and its File System	-	-	-	-
PC9. Working on Command Line Utilities	-	-	-	-
PC10. Working with Links: Working with Symbolic Links, Working with Hard Links	-	-	-	-
PC11. Configuring Permissions, Read, Write, and Execute:	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Preparing the File System for ACLs	-	-	-	-
PC13. Configuring Administrator Tasks with sudo	-	-	-	-
PC14. The in-depth idea about the Linux operating system and its File System	-	-	-	-
PC15. Managing Users and Groups,	-	-	-	-
NOS Total	100	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0901
NOS Name	Fundamentals of Network and Operating System
Sector	Information Technology Sector
Sub-Sector	Networking and Cybersecurity
Occupation	Cyber Security
NSQF Level	3
Credits	3
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

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NIE/ITS/N0901: Operating System Basics

Description

The NOS cover the installation processes of Windows and Linux, including various Linux installation types. It emphasizes using OS commands for file management, system operations, and configuring essential services. Understanding how OS boots, interacts with hardware, and manages services helps in system administration and troubleshooting tasks effectively.

Scope

The scope covers the following :

- The scope of this learning includes gaining hands-on experience with both Windows and Linux OS installation, understanding different installation types, and mastering essential commands for file management and system operations. It also covers configuring and managing services, understanding OS boot processes, hardware interaction, and troubleshooting for efficient system administration.

Elements and Performance Criteria

Install Windows and Linux Operating Systems

To be competent, the user/individual on the job must be able to:

- PC1.** Identify system requirements for Windows and various Linux distributions.
- PC2.** Perform a clean installation of Windows and Linux OS.
- PC3.** Configure initial setup options (language, time zone, users, partitions).

Understand Linux Installation Types

To be competent, the user/individual on the job must be able to:

- PC4.** Compare full installation, minimal installation, and custom installation of Linux.
- PC5.** Select appropriate installation type based on use case (e.g., server vs desktop).
- PC6.** Demonstrate ability to create and configure partitions (manual and guided).

Use Linux Command Line Interface (CLI)

To be competent, the user/individual on the job must be able to:

- PC7.** Execute common Linux commands (navigation, viewing files, managing directories).
- PC8.** Use file manipulation commands (cp, mv, rm, touch, mkdir).
- PC9.** Utilize text editors (nano, vi) for basic file editing.

Perform File and Directory Operations

To be competent, the user/individual on the job must be able to:

- PC10.** Create, rename, and delete files and directories.
- PC11.** Change file permissions and ownership using chmod and chown.
- PC12.** Copy and move files between directories and devices.

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** Learn how to install both Windows and Linux operating systems, explore different Linux installation types, and understand the system boot process and how the OS interacts with hardware.
- KU2.** Build core skills in using operating systems, especially Linux, by practicing essential command-line operations like file handling, system navigation, and permission management.
- KU3.** Understand the purpose of background services in an OS, such as networking and automation tools, and learn how they support system operations and user needs.
- KU4.** Develop the ability to set up, configure, and maintain system services to ensure smooth and secure user access, including monitoring and troubleshooting service-related issues.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Learn to install and configure Windows and Linux OS, and understand the boot process and hardware interaction.
- GS2.** Master basic Linux commands for file creation, deletion, copying, and transferring.
- GS3.** Understand and configure essential system services for user accessibility.
- GS4.** Develop skills to troubleshoot installation issues and analyze system operations.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Install Windows and Linux Operating Systems</i>	9	4	-	-
PC1. Identify system requirements for Windows and various Linux distributions.	-	-	-	-
PC2. Perform a clean installation of Windows and Linux OS.	-	-	-	-
PC3. Configure initial setup options (language, time zone, users, partitions).	-	-	-	-
<i>Understand Linux Installation Types</i>	-	-	-	-
PC4. Compare full installation, minimal installation, and custom installation of Linux.	-	-	-	-
PC5. Select appropriate installation type based on use case (e.g., server vs desktop).	-	-	-	-
PC6. Demonstrate ability to create and configure partitions (manual and guided).	-	-	-	-
<i>Use Linux Command Line Interface (CLI)</i>	-	-	-	-
PC7. Execute common Linux commands (navigation, viewing files, managing directories).	-	-	-	-
PC8. Use file manipulation commands (cp, mv, rm, touch, mkdir).	-	-	-	-
PC9. Utilize text editors (nano, vi) for basic file editing.	-	-	-	-
<i>Perform File and Directory Operations</i>	-	-	-	-
PC10. Create, rename, and delete files and directories.	-	-	-	-
PC11. Change file permissions and ownership using chmod and chown.	-	-	-	-
PC12. Copy and move files between directories and devices.	-	-	-	-
NOS Total	9	4	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0901
NOS Name	Operating System Basics
Sector	Information Technology Sector
Sub-Sector	
Occupation	CYBER SECURITY
NSQF Level	5
Credits	1.67
Version	1.0
Next Review Date	NA

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NIE/ITS/N0902: Fundamentals of Cyber Security

Description

Fundamentals of Cyber Security encompass competencies in core cyber security principles, regulations and policies, threat awareness, network and operating system security, access control, incident response, data protection, ethical hacking, and risk management, providing a framework for individuals to understand, implement, and adhere to essential security practices in the field of cyber security.

Scope

The scope covers the following :

- Fundamentals of Cyber Security encompasses the foundational knowledge and skills required to comprehend, implement, and maintain essential security practices, including threat awareness, network and system security, access control, and incident response, to safeguard digital assets and data from cyber threats and vulnerabilities.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Gaining the basic idea of Information Security and Introduction to Ethical Hacking
- PC2.** Overview of Cryptography
- PC3.** OS security and security tools for Windows Operating System
- PC4.** Gaining the idea of Wireless Networks and Wireless Security
- PC5.** Gaining the idea of the Android and iPhone iOS Operating system
- PC6.** Introduction to Web Application & its terminologies and exploring Web Application Protocols
- PC7.** Configuring Browser Security
- PC8.** . Introduction to Social Media & its security
- PC9.** Exploring Security in Digital Payment Systems
- PC10.** Gaining the idea of Various kinds of fraud in banking and Cyber Crime
- PC11.** Introduction to Cyber Law, Cyber Cell, and Cyber Fraud Helpline

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Comprehensive Understanding of Cybersecurity Concepts -Knowledge of information security fundamentals, ethical hacking, cryptography, and OS-level security measures.
- KU2.** Familiarity with Security Tools and Practices -Awareness of tools and techniques for securing Windows OS, wireless networks, mobile operating systems (Android/iOS), and web applications.
- KU3.** Insight into Emerging Threats and Platforms -Understanding of social media security, digital payment systems, and potential threats in modern digital platforms.

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- KU4.** Awareness of Legal and Regulatory Frameworks- Basic knowledge of cyber crimes, cyber law, fraud detection in banking, and available cybercrime redressal mechanisms such as helplines and cyber cells.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to understand core principles of information security, ethical hacking, cryptography, and OS-level protection across platforms.
- GS2.** Skilled in using cybersecurity tools for securing operating systems, web applications, wireless networks, and mobile platforms.
- GS3.** Knowledge of secure digital payment systems, social media safety, and online fraud prevention.
- GS4.** Familiarity with cyber laws, cybercrime reporting mechanisms, and the role of cyber cells and helplines in ensuring digital justice.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	100	60	-	-
PC1. Gaining the basic idea of Information Security and Introduction to Ethical Hacking	-	-	-	-
PC2. Overview of Cryptography	-	-	-	-
PC3. OS security and security tools for Windows Operating System	-	-	-	-
PC4. Gaining the idea of Wireless Networks and Wireless Security	-	-	-	-
PC5. Gaining the idea of the Android and iPhone iOS Operating system	-	-	-	-
PC6. Introduction to Web Application & its terminologies and exploring Web Application Protocols	-	-	-	-
PC7. Configuring Browser Security	-	-	-	-
PC8. . Introduction to Social Media & its security	-	-	-	-
PC9. Exploring Security in Digital Payment Systems	-	-	-	-
PC10. Gaining the idea of Various kinds of fraud in banking and Cyber Crime	-	-	-	-
PC11. Introduction to Cyber Law, Cyber Cell, and Cyber Fraud Helpline	-	-	-	-
NOS Total	100	60	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0902
NOS Name	Fundamentals of Cyber Security
Sector	Information Technology Sector
Sub-Sector	Networking and Cybersecurity
Occupation	Cyber Security
NSQF Level	3
Credits	5
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

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NIE/ITS/N0902: Computer Network

Description

Develop comprehensive skills in computer networking, including IP addressing, routing, switching, and WAN technologies. Gain hands-on experience configuring network devices, securing networks using ACLs, NAT, VPN, and Layer 2 protections, and implementing network monitoring and logging solutions. Learn to troubleshoot network issues and ensure secure, reliable communication across various protocols and platforms.

Scope

The scope covers the following :

- This module equips learners with practical skills in designing, configuring, and managing both LAN and WAN networks using IPv4 and IPv6. It covers essential topics such as routing, switching, VLANs, NAT, ACLs, VPNs, and network security. Students also gain hands-on experience with device configuration, network monitoring, and troubleshooting to ensure secure and efficient network operations.

Elements and Performance Criteria

Design and Configure a Functional Network

To be competent, the user/individual on the job must be able to:

- PC1.** Identify and explain types of networks, topologies, and models (OSI, TCP/IP).
- PC2.** Design small LAN/WAN topologies and assign IP addressing (IPv4/IPv6) including subnetting using VLSM and FLISM.
- PC3.** Configure network devices using CLI and verify connectivity using protocols such as Telnet, SSH, CDP, and LLDP.

Implement and Troubleshoot Routing & Switching

To be competent, the user/individual on the job must be able to:

- PC4.** Configure static, default, and dynamic routing protocols (RIP, EIGRP, OSPF, IPv6 variants).
- PC5.** Implement Layer 2 switching, VLANs, trunking protocols, and mitigate threats using port-security and ACLs.
- PC6.** Analyze and troubleshoot routing tables, protocol operations, and switching behaviors.

Manage WAN and Network Security Features

To be competent, the user/individual on the job must be able to:

- PC7.** Configure WAN protocols like PPP, HDLC, and Frame Relay.
- PC8.** Implement NAT types, troubleshoot connectivity, and ensure secure Internet access.
- PC9.** Deploy FHRP protocols (HSRP, VRRP, GLBP) and use tracking for redundancy.

Apply Network Monitoring and Synchronization Tools

To be competent, the user/individual on the job must be able to:

- PC10.** Install and configure SNMP and logging servers; set up network devices as agents.
- PC11.** Configure NTP on routers and synchronize time across the network.

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PC12. Deploy site-to-site VPNs to secure data transmission and verify their functionality.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand different network types and topologies, as well as key protocols like TCP/IP and OSI. Learn how to configure IP addresses (IPv4/IPv6), apply subnetting, and set up network devices using the command line. Get hands-on experience with routing protocols and NAT/ACL configurations.
- KU2.** Learn to configure Layer 2 switches, VLANs, and trunking. Develop security skills by implementing port security and managing ACLs to control network traffic and mitigate threats.
- KU3.** Gain knowledge in configuring WAN protocols such as PPP, HDLC, and Frame Relay. Set up secure remote access methods like Telnet, SSH, and VPNs to ensure safe data transmission.
- KU4.** Learn how to install and use network monitoring tools, set up logging servers, and troubleshoot common network issues related to routing, NAT, and device configurations to maintain a secure, efficient network.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Develop problem-solving skills by configuring network devices, IP addressing, routing protocols, and troubleshooting issues such as NAT, routing, and network connectivity problems.
- GS2.** Gain knowledge of networking protocols like TCP/IP, OSPF, RIP, and VLANs, along with understanding network security concepts like ACLs, VPNs, and port security.
- GS3.** Implement security measures such as ACLs, port security, and VPNs. Configure SNMP, logging servers, and network monitoring tools to track and ensure secure network operations.
- GS4.** Configure, verify, and troubleshoot routing protocols, VLANs, and switch settings (Layer 2/Layer 3). Gain hands-on experience with advanced features like OSPF, EIGRP, and NAT.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Design and Configure a Functional Network</i>	55	27	-	-
PC1. Identify and explain types of networks, topologies, and models (OSI, TCP/IP).	-	-	-	-
PC2. Design small LAN/WAN topologies and assign IP addressing (IPv4/IPv6) including subnetting using VLSM and FLSM.	-	-	-	-
PC3. Configure network devices using CLI and verify connectivity using protocols such as Telnet, SSH, CDP, and LLDP.	-	-	-	-
<i>Implement and Troubleshoot Routing & Switching</i>	-	-	-	-
PC4. Configure static, default, and dynamic routing protocols (RIP, EIGRP, OSPF, IPv6 variants).	-	-	-	-
PC5. Implement Layer 2 switching, VLANs, trunking protocols, and mitigate threats using port-security and ACLs.	-	-	-	-
PC6. Analyze and troubleshoot routing tables, protocol operations, and switching behaviors.	-	-	-	-
<i>Manage WAN and Network Security Features</i>	-	-	-	-
PC7. Configure WAN protocols like PPP, HDLC, and Frame Relay.	-	-	-	-
PC8. Implement NAT types, troubleshoot connectivity, and ensure secure Internet access.	-	-	-	-
PC9. Deploy FHRP protocols (HSRP, VRRP, GLBP) and use tracking for redundancy.	-	-	-	-
<i>Apply Network Monitoring and Synchronization Tools</i>	-	-	-	-
PC10. Install and configure SNMP and logging servers; set up network devices as agents.	-	-	-	-
PC11. Configure NTP on routers and synchronize time across the network.	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Deploy site-to-site VPNs to secure data transmission and verify their functionality.	-	-	-	-
NOS Total	55	27	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N0902
NOS Name	Computer Network
Sector	Information Technology Sector
Sub-Sector	
Occupation	CYBER SECURITY
NSQF Level	5
Credits	4.1
Version	1.0
Next Review Date	NA

Qualification Pack

NIE/SSC/N0918: Implementation of Cyber Security Project

Description

Cyber Security Assistant, contribution to projects such as network vulnerability assessments, phishing awareness simulations, and log monitoring for threat detection. I also supported the development of cybersecurity policies to strengthen organizational security. These projects enhanced my skills in threat analysis, incident response, and security awareness.

Scope

The scope covers the following :

- The Cyber Security Assistant supports the implementation and monitoring of security measures to protect computer systems, networks, and data. Their scope includes assisting with threat detection, vulnerability assessments, incident response, and user awareness activities. They help enforce security policies, manage access controls, and ensure compliance with organizational and regulatory standards. The role is crucial in maintaining the confidentiality, integrity, and availability of information systems.

Elements and Performance Criteria

Assist in Monitoring Security Systems

To be competent, the user/individual on the job must be able to:

- PC1.** Monitor system alerts and logs for unusual or unauthorized activity.
- PC2.** Report identified threats or anomalies to the security team.

Support Threat Identification and Response

To be competent, the user/individual on the job must be able to:

- PC3.** Assist in identifying common cyber threats such as phishing, malware, and unauthorized access.
- PC4.** Follow incident response procedures under supervision.

Assist in Conducting Security Assessments

To be competent, the user/individual on the job must be able to:

- PC5.** Participate in vulnerability scans and basic risk assessments.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand basic cybersecurity concepts, common threats (e.g., phishing, malware), and the importance of data confidentiality, integrity, and availability.
- KU2.** Familiar with antivirus software, firewalls, SIEM tools, and basic networking components like IPs, protocols, and ports.
- KU3.** Knowledge of cybersecurity policies, data protection laws (e.g., GDPR), and ethical responsibilities in handling sensitive data.

Qualification Pack

KU4. Capable of assisting in documentation, reporting, user awareness training, and following proper procedures for incident identification and escalation.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Able to work collaboratively and communicate technical information clearly to both technical and non-technical audiences
- GS2.** Skilled in identifying security issues, analyzing risks, and ensuring accuracy in monitoring and reporting.
- GS3.** Capable of handling multiple tasks, meeting deadlines, and maintaining well-structured documentation.
- GS4.** Familiar with cybersecurity tools and practices, and adheres to ethical standards and legal compliance in data handling.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Assist in Monitoring Security Systems</i>	-	-	20	20
PC1. Monitor system alerts and logs for unusual or unauthorized activity.	-	-	-	-
PC2. Report identified threats or anomalies to the security team.	-	-	-	-
<i>Support Threat Identification and Response</i>	-	-	-	-
PC3. Assist in identifying common cyber threats such as phishing, malware, and unauthorized access.	-	-	-	-
PC4. Follow incident response procedures under supervision.	-	-	-	-
<i>Assist in Conducting Security Assessments</i>	-	-	-	-
PC5. Participate in vulnerability scans and basic risk assessments.	-	-	-	-
NOS Total	-	-	20	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/SSC/N0918
NOS Name	Implementation of Cyber Security Project
Sector	IT-ITeS
Sub-Sector	
Occupation	Cyber Security
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment of the qualification evaluates candidates to ascertain that they can integrate knowledge, skills and values for carrying out relevant tasks as

per the defined learning outcomes and assessment criteria.

The underlying principle of assessment is fairness and transparency. The evidence of the outcomes and assessment criteria. competence acquired by

the candidate can be obtained by conducting Theory (Online), Practical assessment, Internal assessment, Project/Presentation/ Assignment, Major

Project. The emphasis is on the practical demonstration of skills & knowledge gained by the candidate through the training. Each OUTCOME is assessed

& marked separately. A candidate is required to pass all OUTCOMES individually based on the passing criteria.

About Examination Pattern:

Qualification Pack

1. The question papers for the theory and practical exams are set by the Examination wing (assessor) of NIELIT HQS.
2. The assessor assigns roll number.
3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.
4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.
5. Pass percentage would be 50% marks in each component.
6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).
7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.
8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.

Approved in 28th NSQC Meeting dated 29/03/2023 Cyber Security Assistant (STT) QG-03-IT-00350-2023-V1-NIELIT

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9. The examination will be conducted in English language only.

Quality assurance activities: A pool of questions is created by a subject matter expert and moderated by other SME. Test rules are set beforehand.

Qualification Pack

Random set of questions which are according to syllabus appears which may differ from candidate to candidate. Confidentiality and impartiality are

maintained during all the examination and evaluation processes.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	13
NIE/ITS/N0901.Fundamentals of Network and Operating System	100	30	-	-	130	34
NIE/ITS/N0902.Fundamentals of Cyber Security	100	60	-	-	160	42
NIE/SSC/N0918.Implementation of Cyber Security Project	-	-	20	20	40	11
Total	220	120	20	20	380	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework
NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualifications Framework
OJT	On the Job Training
AA	Assessment Agency
AB	Awarding Body
NCrF	National Credit Framework
NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualification Framework
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Qualification Pack

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Qualification Pack

NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualifications Framework
OJT	On the Job Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

Qualification Pack

Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
National Occupational Standard	NOS define the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology.
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above. https://ncvet.gov.in/sites/default/files/NCVET.pdf
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